SKYWORTH
Your Best Friend of Energy-Saving Solutions

SWH
3/3.6/5/6/8/10/12kW

Single Phase Hybrid Inverter (LV)

MODEL

SWH3000L-S1-G2 | SWH3600L-S1-G2 | SWH5000L-S1-G2 | SWH6000L-S1-G2

SWH8000L-S1-G2 | SWH010KL-S1-G2 | SWH012KL-S1-G2



Higher Power Generation
· Max. 20A DC input current per string
· 200% PV input oversizing

Smart Control & Monitoring
· Smart load control
· Seamless switching to backup <4ms

Superb Safety & Reliability
· AI-driven AFCI [1]
· IP66 ingress protection

Flexible & Adaptable Applications
· 3-in-1 Port: generator & smart loads & on-grid inverter
· Micro-grid operation with PV inverter
· Supports on- and off-grid parallel operation of up to 6 units

1: Optional functions or devices are purchased separately.

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Technical Data	SWH3000L-S1-G2	SWH3600L-S1-G2	SWH5000L-S1-G2	SWH6000L-S1-G2	SWH8000L-S1-G2	SWH010KL-S1-G2	SWH012KL-S1-G2
Battery Side							
Battery Type*1				Li-Ion / Lead-acid			
Nominal Battery Voltage (V)				48			
Battery Voltage Range (V)				40 ~ 60			
Start-up Voltage (V)				44.2			
Number of Battery Input				1			
Max. Continuous Charging Current (A)	70	90	120	140	160	200	240
Max. Continuous Discharging Current (A)	70	90	120	140	160	200	240
Max. Charging Power (W)	3000*2	3600*2	5000*2	6000*2	8000*3	10000*3	12000*3
Max. Discharging Power (W)	3300*2	3960*2	5000*2	6600*2	8800*3	11000*3	13200*3
PV String Input Data							
Max. Input Power (W)	6000	7200	10000	12000	16000	20000	24000
Max. Input Voltage (V)*4				600			
MPPT Operating Voltage Range (V)*5				60 ~ 550			
Start-up Voltage (V)				58			
Nominal Input Voltage (V)				360			
Max. Input Current per MPPT (A)	20	20	20	20	32 / 16*6	32 / 32*6	32 / 32*6
Max. Short Circuit Current per MPPT (A)	26	26	26	26	48 / 24	48 / 48	48 / 48
Number of MPP Trackers	1	2	2	2	2	2	2
Number of Strings per MPPT	1	1	1	1	2 / 1	2/ 2	2/ 2
AC Output Data (On-grid)							
Nominal Output Power (W)	3000	3600	5000	6000	8000	10000	12000
Nominal Apparent Power Output to Utility Grid (VA)	3000	3600	5000	6000	8000	10000	12000
Max. AC Active Power (W)*7*8	3300	3960*9	5500	6600	8800	11000	13200
Max. Apparent Power Output to Utility Grid (VA)*7*8	3300	3960*9	5500	6600	8800	11000	13200
Max. Apparent Power from Utility Grid (VA)	7040	7040	8000	8800	16500	16500	16500
Nominal Output Voltage (V)				220 / 230 / 240			
Output Voltage Range (V)				170 ~ 280			
Nominal AC Grid Frequency (Hz)				50 / 60			
AC Grid Frequency Range (Hz)				45 ~ 55 / 55 ~ 65			
Max. AC Current Output to Utility Grid (A)	15	18	25	30	40	50	60
Max. AC Current From Utility Grid (A)	32	32	40	70	75	75	75
Power Factor				~1 (Adjustable from 0.8 leading to 0.8 lagging)			
Max. Total Harmonic Distortion				<3%			
AC Output Data (Back-up)							
Back-up Nominal Apparent Power (VA)	3000	3600	5000	6000	8000	10000	12000
Max. Output Apparent Power without Grid (VA)	3300 (6000, 10s)	3960 (7200, 10s)	5500 (10000, 10s)	6600 (12000, 10s)	8800 (16000, 10s)	11000 (20000, 10s)	13200 (24000, 10s)
Max. Output Apparent Power with Grid (VA)	7040	7040	8000	8800	16500	16500	16500
Max. Output Current without Grid (A)	15	18	25	30	40	50	60
Max. Output Current with Grid (A)	32	32	40	40	75	75	75
Nominal Output Voltage (V)				220 / 230 / 240			
Nominal Output Frequency (Hz)				50 / 60			
Output THDv (@Linear Load)				<3%			
Efficiency							
Max. Efficiency				97.6%			
European Efficiency				96.2%			
Max. Battery to AC Efficiency				95.5%			
MPPT Efficiency				99.9%			
Protection							
DC Switch				●			
DC Surge Protection				Type II			
AC Surge Protection				Type II			
AFCI				○			
Rapid Shutdown				○			
Remote Shutdown				●			
General Data							
Operating Temperature Range (°C)				-35 ~ +60			
Relative Humidity				0 ~ 95%			
Max. Operating Altitude (m)				3000			
Cooling Method		Natural Convection				Smart Fan Cooling	
User Interface				LCD, WLAN + APP			
Communication with BMS				CAN			
Communication				RS485, WiFI + LAN + Bluetooth			
Communication Protocols				Modbus-RTU, Modbus-TCP			
Weight (kg)	14.5	15.5	15.5	15.5	27.0	29.0	29.0
Dimension (W × H × D mm)		560 × 415 × 204				560 × 444.5 × 226	
Topology				Non-isolated			
Ingress Protection Rating				IP66			
Mounting Method				Wall Mounted			

● Integrated ○ Optional

*1: The Li-Ion battery usually contain two mainstream type: LFP and Ternary Lithium battery.

*2: When the PV input voltage is higher than 500V, the battery charging and discharging power will be gradually limited, and the power limitation will be lifted after the input voltage is lowered.

*3: When the PV input voltage is higher than 490V, the battery charging and discharging power will be gradually limited, and the power limitation will be lifted after the input voltage is lowered.

*4: When the input voltage is 560V-600V, the inverter will enter standby mode. The inverter will return to normal operation state when the voltage returns to the MPPT working voltage range.

*5: Please refer to the user manual for the MPPT Voltage Range at Nominal Power.

*6: The maximum input current per string is 16A. Or For the MPPT with two strings, the current of each string is 16A.

*7: For Sri Lanka, Max. Output Power (kW) is 3.0kW for SWH3000L-S1-G2, 3.6kW for SWH3000L-S1-G2, 5.0kW for SWH5000L-S1-G2,6.0kW for SWH6000L-S1-G2, 8.0kW for SWH6000L-S1-G2, 10.0kW for SWH010KL-S1-G2, and 12.0kW for SWH012KL-S1-G2.

*8: For Brazil and Chile, the max. AC output power is Pn, such as the max. AC output power of SWH8000L-S1-G2 is 8000W (VA).

*9: For Jordan, Max. AC Active Power and Max. Apparent Power Output to Utility Grid are 3.6kW/kVA for SWH3600L-S1-G2.

*10: For Jordan, Max. AC Current Output to Utility Grid is 15.7A for SWH3600L-S1-G2.

*: Please visit SKYWORTH website for the latest certificates.

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